

# NurseBridge

Voice intake with context for the nurse

AssemblyAI Voice Agent Hackathon

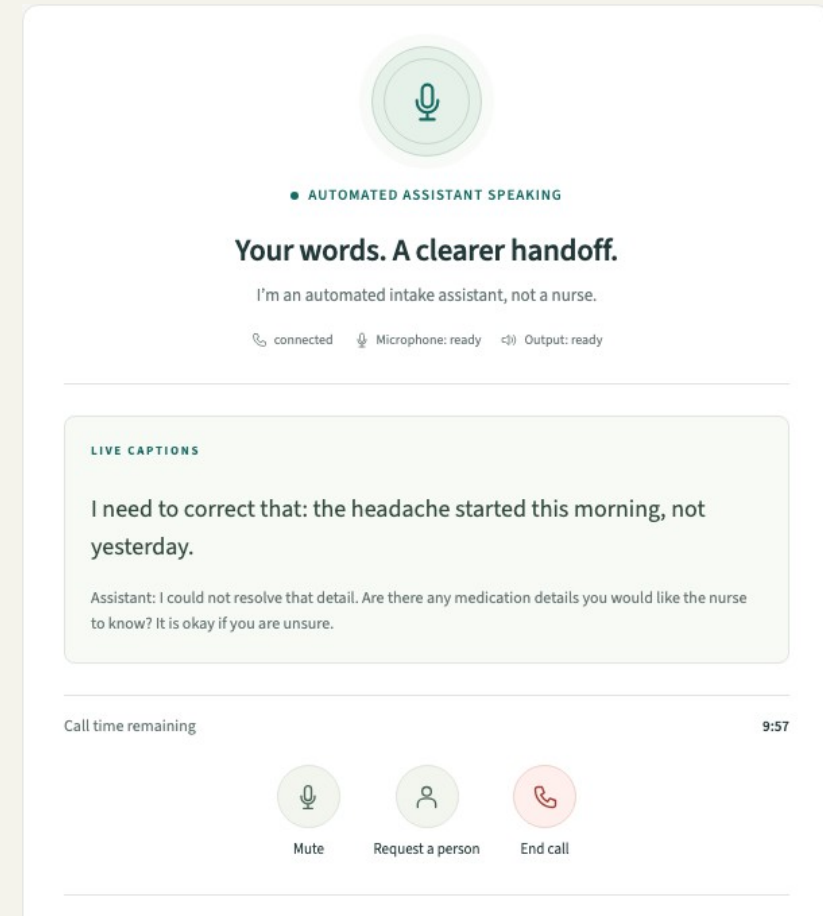
Simulation only. Fictional information. Not for medical care.

# The intake handoff

**A caller has a story.  
The nurse needs usable context.**

Details can change during a conversation. A useful intake draft needs to retain corrections and uncertainty.

NurseBridge explores collecting that context while the caller waits for a nurse.



Mock replay interface

# Caller to nurse

## 1 Join the queue

Human access remains available.

## 2 Share a fictional story

The assistant collects intake details.

## 3 Review the draft

A nurse can inspect source words.

## 4 Take over the call

Automated forwarding stops.

BROWSER CALL · B574577A

Transcript replay

**Caller 577A**

Intake interrupted · Arrived 01:41 AM · Awaiting nurse review

Human audio connected

Automated provider forwarding is stopped.

Connected

Caller online

Your microphone: ready

Your output: ready

Connection: connected

Overall time: 9:56

**Intake draft**

7 fields captured

Patient-reported information. Evidence shows provenance, not medical validation.

REASON FOR CALLING

**a headache**

Caller's wording: "I am calling about a headache that started yesterday afternoon."

"I am calling about a headache that started yesterday afternoon."

Patient reported

Fictional mock replay with browser human audio

# Corrections retain source words

“The headache started this morning, not yesterday.”

The current draft changes. The original words and correction history remain available for review.

The screenshot displays a medical call interface with two main panels. The left panel, titled 'BROWSER CALL · B574577A', shows the call status as 'Intake in progress' and 'Awaiting nurse review'. It features a 'Take over call' button and a status bar indicating 'Caller online', 'Your microphone: idle', 'Your output: idle', 'Connection: connected', and 'Overall time: 9:58'. The 'Intake draft' section shows 'Patient-reported information. Evidence shows provenance, not medical validation.' and '6 fields captured'. The 'REASON FOR CALLING' section shows 'a headache' and 'Caller's wording: "I am calling about a headache that started yesterday afternoon."' with a 'View supporting words' link. The 'ONSET & DURATION' section shows 'I need to correct that: the headache started this morning, not yesterday.' and 'Nurse review pending · Not patient-confirmed' with an 'Edit with evidence' link. The right panel, titled 'Transcript Revisions Timeline', shows a list of revisions. The first revision, 'Revision 1', shows the original text 'I am calling about a headache that started yesterday afternoon:' and the correction 'I need to correct that: the headache started this morning, not yesterday.' with a 'View supporting words' link. The second revision, 'Revision 2', shows the current draft text 'I am calling about a headache that started yesterday afternoon:' and the correction 'I need to correct that: the headache started this morning, not yesterday.' with a 'View supporting words' link.

BROWSER CALL · B574577A

Transcript replay

Caller 577A

Intake in progress Arrived 01:41 AM · Awaiting nurse review

Assistant speaking  
You can join before intake is complete.

Take over call

Caller online Your microphone: idle Your output: idle Connection: connected Overall time: 9:58

Intake draft 6 fields captured  
Patient-reported information. Evidence shows provenance, not medical validation.

REASON FOR CALLING Patient reported  
a headache  
Caller's wording: "I am calling about a headache that started yesterday afternoon."  
"I am calling about a headache that started yesterday afternoon."  
Nurse review pending · Not patient-confirmed Edit with evidence

ONSET & DURATION Patient reported  
I need to correct that: the headache started this morning, not yesterday.  
"I need to correct that: the headache started this morning, not yesterday."  
Nurse review pending · Not patient-confirmed Edit with evidence

Transcript Revisions Timeline

Onset & duration Revision 2  
I am calling about a headache that started yesterday afternoon:  
I need to correct that: the headache started this morning, not yesterday.  
01:41 AM · Automated extraction · review required  
View supporting words

Uncertainties Revision 1  
I have not checked my temperature.  
01:41 AM · Automated extraction · review required  
View supporting words

Reported symptoms Revision 1  
No other symptoms.  
01:41 AM · Automated extraction · review required  
View supporting words

Caller-described severity Revision 1  
I would describe it as a six out of ten.  
01:41 AM · Automated extraction · review required  
View supporting words

Unmeasured and uncertain details remain explicit. The draft awaits nurse review.

# Technology and control

## Conversation

### AssemblyAI Voice Agent

Speech, turn-taking and agent audio

### Nebius Nemotron-3.5-Lightning

Conversation model and independent structured extraction

## Application authority

### Cloudflare Workers

Next.js through OpenNext

### Durable Object

Consent, evidence revisions and nurse takeover

D1 stores projections. R2 holds private exports.

Human conversation bypasses the automated providers after takeover.

# Evidence and its limits

165

Unit tests pass

173

Workers tests pass

29

Browser tests pass

29 Sep 2026 local regression run. 19 tooling tests also passed.  
The regression run skipped one separate live-provider test.

## Local real-provider proof

29 Sep 2026: finalized transcript, evidence-linked fact and two-way browser audio using recorded microphones.

## Remaining release gates

Hosted operation, physical-device audio, provider recording controls and real telephone calls.

# Proposed users and next validation

## Proposed users

**Nurses who need caller context before joining a conversation.**

## Business hypothesis

A paid workspace for nurse-line operators, if interviews and workflow trials establish value.

## Next validation

Nurse interviews and fictional case reviews

Two physical devices and audible handoff

Provider recording controls and hosted acceptance

Live telephone call for the optional phone path